

9-22

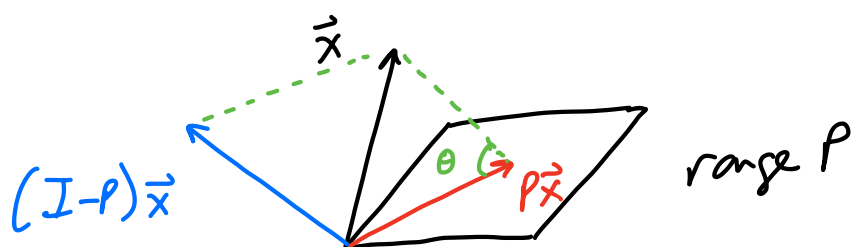
(See Chapter 4 of Strang)

Theorem 8.1 (Properties of Projections) Let P be a projection matrix so that $P^2 = P$. Then

- i. The complement of P , $I - P$ is also a projection
- ii. P acts as the identity on range P
- iii. $\text{null } P = \text{range}(I - P)$ and $\text{null}(I - P) = \text{range } P$

Proof: i. $(I - P)^2 = I - 2P + P^2 = I - P$

ii., iii. (Exercise)



As a result 8.1.ii. we say that P is a projection onto range P and that $P\vec{x}$ is the projection of $\vec{x}$ onto range P .

Corollary 8.2 If $P \in F^{n \times n}$ is a projection then

$$F^n = \text{range } P \oplus \text{null } P.$$

Proof: As $\vec{x} = P\vec{x} + (I - P)\vec{x}$ this is mostly

follows from property 8.1.iii. We only need to show that $\text{range } P \cap \text{null } P = \emptyset$.
Letting $\vec{x} \in \text{range } P \cap \text{null } P$ we have that
by 8.1.ii., $P\vec{x} = \vec{x}$ and $P\vec{x} = \vec{0}$.
Thus $\vec{x} = \vec{0}$ as desired.

Def 8.3 $P \in \mathbb{R}^{n \times n}$ is said to be an orthogonal projection onto $\text{range } P$ iff $P^2 = P$ and $P^T = P$
(P is a projection that is symmetric or self-adjoint)

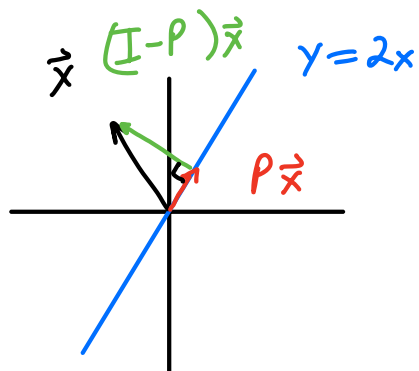
Theorem 8.4 (Properties of Orthogonal Projections)

If $P \in \mathbb{R}^{n \times n}$ is an orthogonal projection onto $W (= \text{range } P)$, then

- i. $\text{range } P \perp \text{null } P$ ($\text{null } P = (\text{range } P)^\perp$)
- ii. If P' is another orthogonal projection onto W , then $P' = P$ (uniqueness)
- iii. $I - P$ is the (unique) orthogonal projection onto $\text{null } P$

Exercise:

- 1) Find the matrix representation of the orthogonal projection onto the line $y=2x$



Let $T \in \mathcal{L}(\mathbb{R}^2, \mathbb{R}^2)$ be such that

$$T \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \text{ and } T \begin{bmatrix} 2 \\ -1 \end{bmatrix} = \vec{0}.$$

Now

$$[T]_{\{e_i\}} = \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}^{-1}$$

$$= \frac{1}{5} \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$= \frac{1}{5} \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}$$

Denoting $\vec{v} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ we have that

$$P = [T]_{\{e_i\}} = \frac{\vec{v} \vec{v}^T}{\vec{v}^T \vec{v}}$$

Def 8.5 Let $A \in \mathbb{R}^{n \times m}$ where $m \leq n$ have lin. ind. columns. ($\text{rank } A = m$) We will denote P_A as the orthogonal projection onto range A .

Theorem 8.6 letting A be as above
($A \in \mathbb{R}^{n \times m}$ with $\text{rank } A = m < n$) we have that

$$P_A = A(A^T A)^{-1} A^T \quad (P_A = I \text{ if } m=n)$$

Caution: $(A^T A)^{-1} \neq A^{-1}(A^T)^{-1}$ as A is not invertible ($m < n$)

Proof: let $A \in \mathbb{R}^{n \times m}$ have rank $m < n$. Now let P_A be such that $P_A(A\vec{x}) = A\vec{x}$ and $\text{null } P_A$ is orthogonal to $\text{range } P_A$. Since $\text{range } P_A = \text{range } A$, we have that

$$\begin{aligned} \text{null } P_A &= (\text{range } P_A)^\perp = (\text{range } A)^\perp \\ &= \text{null } A^T \end{aligned}$$

and thus we set $P_A = M \cdot A^T$ where $M \in \mathbb{R}^{n \times m}$ and $\text{null } M = \emptyset$. For any $\vec{x} \in \mathbb{R}^m$ we have that $A\vec{x} \in \text{range } A$ and thus

$$A\vec{x} = P_A(A\vec{x}) = M(A^T A)\vec{x}$$

Hence $A = M(A^T A)$ and thus $M = A(A^T A)^{-1}$ as $A^T A \in \mathbb{R}^{m \times m}$ is invertible as desired.

Exercise:

2) Find the orthogonal projection onto the plane $W = \{(x_1, x_2, x_3) \in \mathbb{R}^3 : 2x_1 - x_2 - x_3 = 0\}$

need $A \in \mathbb{R}^{3 \times 2}$ s.t. $W = \text{range } A$

Setting $x_2 = 2x_1 - x_3$ we have that

$$W = \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} \right)$$

and setting $A = \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 1 \end{bmatrix}$ we have that

$$P_A (= P_W) = A(A^T A)^{-1} A^T$$

$$= \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 1 \end{bmatrix} \left(\begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 1 & 0 \\ 2 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$= \frac{1}{6} \begin{bmatrix} 2 & 2 & 2 \\ 2 & 5 & -1 \\ 2 & -1 & 5 \end{bmatrix}$$

$$\text{check: } P_W \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 2 & 2 & 2 \\ 2 & 5 & -1 \\ 2 & -1 & 5 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} = \frac{1}{6} \begin{bmatrix} 6 \\ 12 \\ 0 \end{bmatrix}$$

end of exam 1 stuff

Lemma 8.7 let ∇ denote the gradient and let $A \in \mathbb{R}^{n \times n}$. Then $\nabla \langle \vec{x}, A\vec{x} \rangle = (A + A^T)\vec{x}$

If $A^T = A$, then $\nabla \langle \vec{x}, A\vec{x} \rangle = 2A\vec{x}$

Proof idea: let $\vec{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ and denote by $A_{i,j}$

the i,j th component of A . Then

$$\langle \vec{x}, A\vec{x} \rangle = \sum_i \sum_j A_{i,j} x_i x_j$$

$$= \sum_i A_{i,i} x_i^2 + \sum_{i \neq j} A_{i,j} x_i x_j$$

$$\text{and } \frac{\partial}{\partial x_k} \langle \vec{x}, A\vec{x} \rangle = 2A_{k,k} x_k + \sum_{i \neq k} (A_{i,k} + A_{k,i}) x_i$$

$$= \sum_i (A_{i,k} + A_{i,k}^T) x_i$$

Theorem 8.8 let $A \in \mathbb{R}^{n \times m}$ have rank m

and assume that $\vec{b} \notin \text{range } A$. Now setting

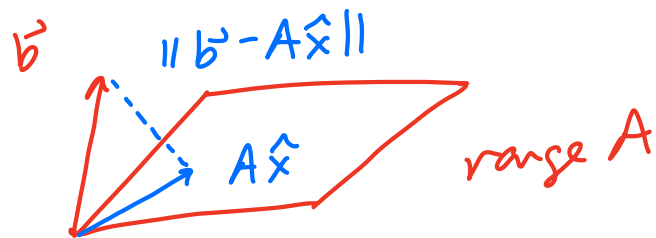
$\hat{x} = (A^T A)^{-1} A^T \vec{b}$ we have that

$$A\hat{x} = \min_{\vec{y} \in \text{range } A} \|\vec{b} - \vec{y}\|$$

or i.e. $A\hat{x}$ minimized $\|\vec{b} - A\vec{x}\|$ in ℓ^2 norm.

Here we call $\hat{x}$ the least-squares solution

to $A\vec{x} = \vec{b}$.



Proof: Since $\nabla \|\vec{b} - A\vec{x}\|^2 = \nabla \langle \vec{b} - A\vec{x}, \vec{b} - A\vec{x} \rangle$

$$= \nabla \left[\|\vec{b}\|^2 - 2 \langle A\vec{x}, \vec{b} \rangle + \langle A\vec{x}, A\vec{x} \rangle \right]$$

$$\nabla \langle \vec{x}, \vec{c} \rangle = \vec{c}$$

$$= \nabla \left[\langle \vec{x}, A^T A \vec{x} \rangle - 2 \langle \vec{x}, A^T \vec{b} \rangle \right]$$

$$= 2 A^T A \vec{x} - 2 A^T \vec{b}$$

and setting $\nabla \|\vec{b} - A\vec{x}\|^2 = \vec{0}$ we get

$$A^T A \vec{x} = A^T \vec{b} \quad \text{or} \quad \vec{x} = (A^T A)^{-1} A^T \vec{b}.$$

Terminology: let $W = \text{range } A$ so that

$\hat{x} = (A^T A)^{-1} A^T \vec{b}$ is the least-squares solution to $A\vec{x} = \vec{b}$. Then we say that $A\hat{x}$ is the orthogonal projection of $\vec{b}$ onto W and $A\hat{x}$

is the vector in W closest to $\vec{b}$ (in the ℓ^2 norm)

$$A\hat{x} = P_A \vec{b}$$

Exercise:

3) Find the vector in $\text{Span}\left(\begin{bmatrix} 1 \\ -1 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix}\right)$ that is closest (in ℓ^2 norm) to $\begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$

Setting $A = \begin{bmatrix} 1 & 1 \\ -1 & -2 \\ 3 & -1 \end{bmatrix}$ we compute

$$P_A = A(A^T A)^{-1} A^T$$

$$= \begin{bmatrix} 1 & 1 \\ -1 & -2 \\ 3 & -1 \end{bmatrix} \left(\begin{bmatrix} 1 & -1 & 3 \\ 1 & -2 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -2 \\ 3 & -1 \end{bmatrix} \right)^{-1} \begin{bmatrix} 1 & -1 & 3 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ -1 & -2 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} 11 & 0 \\ 0 & 6 \end{bmatrix}^{-1} \begin{bmatrix} 1 & -1 & 3 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \frac{1}{66} \begin{bmatrix} 17 & -28 & 7 \\ -28 & 50 & 4 \\ 7 & 4 & 65 \end{bmatrix}$$

Thus the vector in the plane that is closest to $\begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$ is given by

$$\frac{1}{66} \begin{bmatrix} 17 & -28 & 7 \\ -28 & 50 & 4 \\ 7 & 4 & 65 \end{bmatrix} \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{66} \begin{bmatrix} 19 \\ -8 \\ 101 \end{bmatrix}$$